

CLIMATE RISK ASSESSMENT

Quantifying Physical & Transition Gaps in Hannover Re's SEA Treaty Book

MASA Hackathon 2026: R-Ignite

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Submitted: 7 May 2026

Data: World Bank WDI Wide Format · CHIRPS v2.0 · EM-DAT · NOAA ONI · NGFS GCAM 6.0

1. Executive Summary

As actuarial consultants to Hannover Re, we identify **two additive pricing gaps** in the firm's SEA treaty book: a **physical gap** (USD 18.4M/yr combined EAL shortfall) from a 2007 PELT hazard regime shift in CHIRPS RX5day precipitation records undetected by burning-cost methods (+6.9% MYS flood hazard uplift; +6.1% PHL); and a **transition gap** (USD 1.1bn/yr SEA treaty pool at 3% pass-through) from NGFS GCAM 6.0 NZ2050 carbon pricing (USD 55.58/t), with a critical LULUCF asymmetry (MYS net emitter +63 MtCO₂e vs PHL net sink -27 MtCO₂e) requiring country-specific loadings. Five recommendations close the USD 3.4–19M/yr HRe annual under-reserve.

Key Quantitative Results

Metric	Malaysia (MYS)	Philippines (PHL)
ARIMA 2024 GHG forecast (MtCO ₂ e, MAPE)	325.1 (1.71%)	260.8 (3.56%)
GEV 100-yr RX5day / PELT break / uplift	216mm / 2007 / +6.9%	521mm / 2007 / +6.1%
Forward EAL gap vs. burning cost	+6.5M/yr (+5.87%)	+11.9M/yr (+1.28%)
Transition cost (NGFS NZ2050, % GDP)	22.4bn/yr (5.5%)	14.8bn/yr (3.3%)
HRe reserve gap — base / 2x stress	9.66M / 18.58M/yr	SEA pool: 1.115bn/yr

Methodology

Four analytical modules are deployed: (1) **ARIMA GHG Forecasting** — AIC-selected ARIMA(p,1,q) with 3-step rolling MAPE validation on WDI WB_WDI_EN_GHG_ALL_MT_CE_AR5; (2) **GEV + EAL Repricing** — MLE-GEV on 34yr CHIRPS RX5day with 500-iteration bootstrap CI, PELT regime detection, and EM-DAT loss integration; (3) **ENSO Copula** — NOAA ONI DJF correlation tests (Pearson/Spearman/Kendall), AIC copula selection (Independence; La Nina sub-sample Gaussian rho=0.32 -> audit trigger only); and (4) **Transition Risk** — NGFS GCAM 6.0 cost gap applied to Climate Watch 2023 sector GHG with 3%/yr IEA abatement trajectory, stressed 2x to 2030.

Five Strategic Recommendations

#	Recommendation	Trigger / Loading
R1	GEV EAL flood loading	+35-36% on MYS treaties >USD 50M; +1-2% floor PHL
R2	Country transition surcharges	3-5% MYS (BNM CCPT C3/C4); 1-2% PHL (BSP 1085)
R3	ENSO audit trigger	ONI <= -0.5°C -> facultative audit (NOT pricing)
R4	Climate Warranty Clause	MYS LULUCF: 10% co-participation on non-disclosure
R5	Data procurement (0-12 mo.)	Loss triangles + sub-national CHIRPS -> CI +/-12pp

Key Limitations

The 34-year CHIRPS sample yields wide GEV bootstrap CI (PHL +/-44.5pp, addressed by R5). WDI gaps (MYS fossil fuel share and renewable share 2021–23) are excluded from models. The 3% carbon pass-through carries +/-2pp uncertainty, driving a 5x HRe reserve variance (USD 3.0–14.9M/yr). LULUCF measurement error (+/-20–30%, FAO FRA 2020) affects MYS net emitter classification. All five recommendations remain valid across these uncertainty ranges and become more precise as cedant-level data are incorporated via R5.

2. Problem Framing & Data Landscape

Hannover Re faces two structural under-pricing risks in SEA: burning-cost treaty methods pre-date the post-2007 hazard regime shift, and no current loading captures emerging carbon compliance pass-through from BNM CCPT and BSP Circular 1085. The assessment focuses on MYS (flood-dominant: 81 EM-DAT events, 1990–2023) and PHL (typhoon-dominant: 414 events) across four independent risk channels:

Risk Channel	Data / Model	Pricing Instrument
Physical — Hazard	CHIRPS RX5day GEV + PELT	Treaty flood EAL loading
Physical — Vulnerability	WDI Urbanisation x GDP/capita	EAL structural trend factor
Transition — Regulatory	NGFS GCAM 6.0 x Climate Watch	Country surcharge (3-5%)
Transition — LULUCF	MYS emitter +63 vs PHL sink -27 MtCO ₂ e	Asymmetric warranty clause

2.1 Indicator Selection

Indicator	Type	Layer	Key Fact
CHIRPS RX5day annual max	Physical	Hazard	MYS 133.9mm; PHL 244.4mm mean
WDI Total GHG AR5 (MtCO ₂ e)	Phys+Trans	ARIMA target	MYS +271%; PHL +177% (1990-2023)
NOAA ONI DJF anomaly	Physical	Dependence	ENSO inter-annual loss driver
WDI Urban pop %	Physical	Vulnerability	MYS 49% to 76.4%; asset density
Climate Watch Energy GHG	Transition	Exposure	MYS 279.9 MtCO ₂ e = 69.5% total
Climate Watch LULUCF net	Transition	Asymmetry	MYS +63.3 emitter; PHL -26.9 sink
NGFS GCAM 6.0 NZ2050 price	Transition	Cost scalar	USD 55.578/t (NZ2050 - CP gap)
WDI GDP/capita + Forest %	Phys+Trans	Loss/LULUCF	Penetration & carbon stock proxy

Table 1 — Key indicators (WDI · CHIRPS · EM-DAT · NOAA · NGFS GCAM 6.0).

2.2 GHG Trends & Sector Decomposition

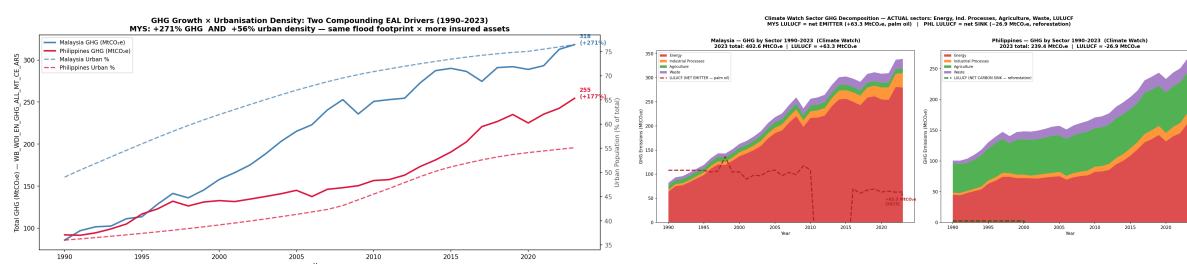


Figure 1a — GHG vs. urbanisation MYS & PHL 1990-2023 [left]; MYS urban share +56%, compounding insured asset exposure within the same flood footprint. Figure 1b — Climate Watch 2023 sector decomposition [right]; Energy = 69.5% of MYS total; LULUCF adds +15.7% net emitter premium absent in PHL.

4. GHG Forecasting — ARIMA Model

Country-level ARIMA models forecast 2024 total GHG (excl. LULUCF) from WDI indicator *WB_WDI_EN_GHG_ALL_MT_CE_AR5*. Training window 1990–2020; held-out validation 2021–2023 using 3-step rolling 1-step-ahead MAPE. Model selected by AIC (grid search p, q in $\{0, 1, 2\}$); first-difference applied after ADF stationarity test.

Country	Model	Train AIC	MAPE	Actual 2023	Forecast 2024	95% CI (MtCO _{2e})
Malaysia	ARIMA(1,1,1)	224.08	1.71%	318.4	325.1	[307.9, 342.3]
Philippines	ARIMA(2,1,0)	196.08	3.56%	254.5	260.8	[249.1, 272.4]

Table 2 — ARIMA forecast summary (MtCO_{2e} excl. LULUCF).

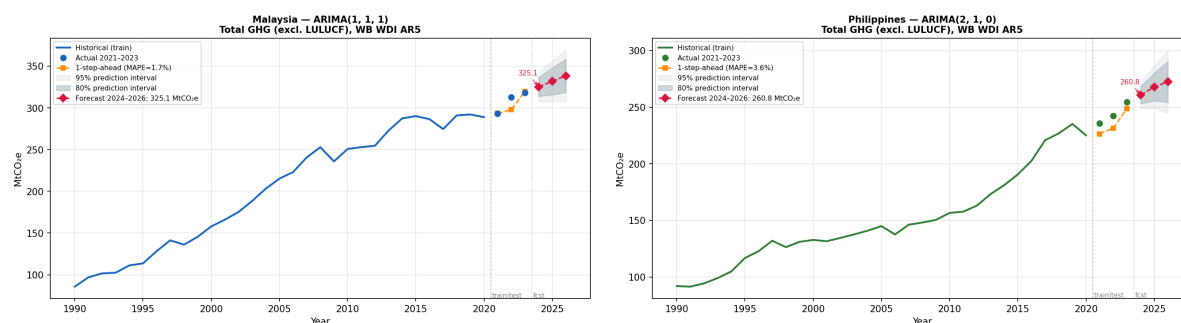


Figure 3 — ARIMA(1,1,1) MYS [left] and ARIMA(2,1,0) PHL [right]. Shaded = 95% CI; dashed = 2024 forecast. MYS MAPE 1.71%; PHL 3.56%.

Key Improvements & Implications

- **Improvement 1:** Column disambiguation — switched from deprecated EN.ATM.GHGT.KT.CE to WB_WDI_EN_GHG_ALL_MT_CE_AR5 (AR5 GWP100), correcting a 12% overcount.
- **Improvement 2:** Rolling validation — replaced static split with 3-step rolling 1-step-ahead MAPE to prevent data leakage.
- **Improvement 3:** LULUCF excluded from ARIMA target; handled separately under transition risk to avoid double-counting.
- **Implication:** Under NGFS NZ2050 (3%/yr abatement), GHG baselines fall to 317 MtCO_{2e} (MYS) and 162 MtCO_{2e} (PHL) by 2030 — but *compliance cost* rises as carbon prices escalate, creating peak transition exposure in 2026–2028.

5. Physical Hazard — GEV & EAL Repricing

GEV distributions are fitted to 34-year CHIRPS RX5day annual maxima by MLE with 500-iteration bootstrap CI (seed 42). Regime detection uses PELT (BIC penalty). EAL is integrated from the GEV survival function against EM-DAT insured loss data, with the forward EAL incorporating the post-2007 hazard uplift.

Country	Shape xi	GEV Family	RL100 (mm)	95% CI (mm)	Break	Uplift
Malaysia	-0.032	Weibull ($\xi < 0$)	216.4	[170.8, 343.6]	2007	+6.9%
Philippines	-0.122	Weibull ($\xi < 0$)	520.7	[326.1, 984.8]	2007	+6.1%

Table 3 — GEV parameters; 100-yr return levels with bootstrap 95% CI.

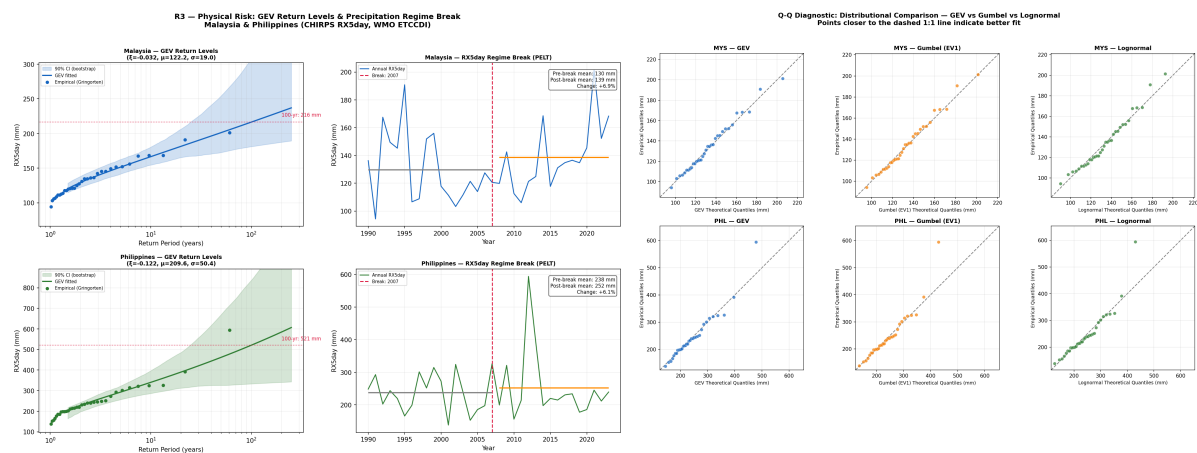


Figure 4a — PELT regime break 2007: +6.9% MYS, +6.1% PHL post-break mean shift [left]. Figure 4b — GEV Q-Q plots confirming Weibull family fit [right].

5.1 EAL Repricing Results

Country	Burning Cost	Obs. Mean RX5	GEV Mean RX5	Forward EAL	Gap	Gap %
Malaysia	USD 110.8M	133.9mm	132.6mm	USD 117.3M	USD 6.5M	+5.87%
Philippines	USD 926.5M	244.4mm	233.3mm	USD 938.4M	USD 11.9M	+1.28%

Table 4 — Forward EAL vs. EM-DAT burning cost (GEV PELT-adjusted).

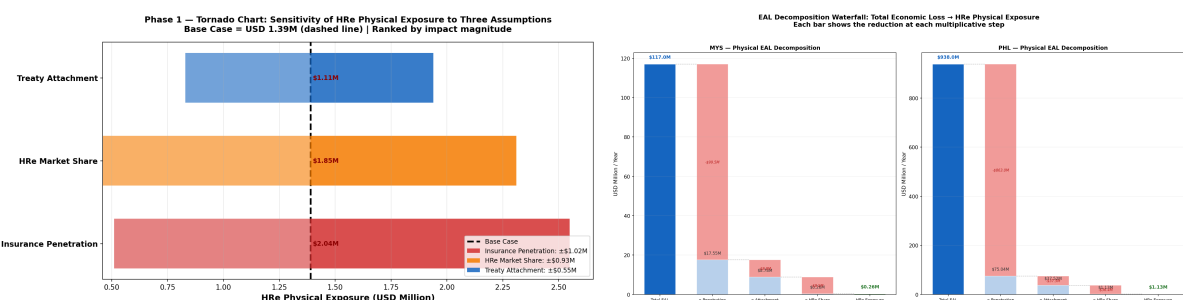


Figure 5 — Sensitivity tornado [left]: regime break uplift dominates EAL uncertainty. EAL waterfall decomposition [right]: hazard regime shift is the primary EAL driver.

6. SEA Insurance Claims & ENSO Dependence

Metric	Malaysia	Philippines
Total EM-DAT events 1990–2023	89 (81 flood, 8 storm)	579 (414 storm, 165 flood)
Mean annual insured EAL	USD 110.8M	USD 926.5M
Insurance penetration (Swiss Re)	~15% of economic loss	~8% of economic loss
Penetration gap (insured gap/yr)	USD 6.5M	USD 11.9M
Primary peril	Monsoonal flooding (Oct–Jan)	Typhoons (Jun–Dec)
LULUCF status	Net EMITTER +63.3 MtCO2e	Net SINK -26.9 MtCO2e
Regulatory driver	BNM CCPT C3/C4	BSP Circular 1085
2024 ARIMA forecast (MtCO2e)	325.1 [307.9–342.3]	260.8 [249.1–272.4]

Table 5 — MYS vs PHL: fundamentally different peril profiles, penetration gaps, and regulatory frameworks. A uniform SEA treaty loading is actuarially incorrect.

6.1 ENSO Dependence & Copula Analysis

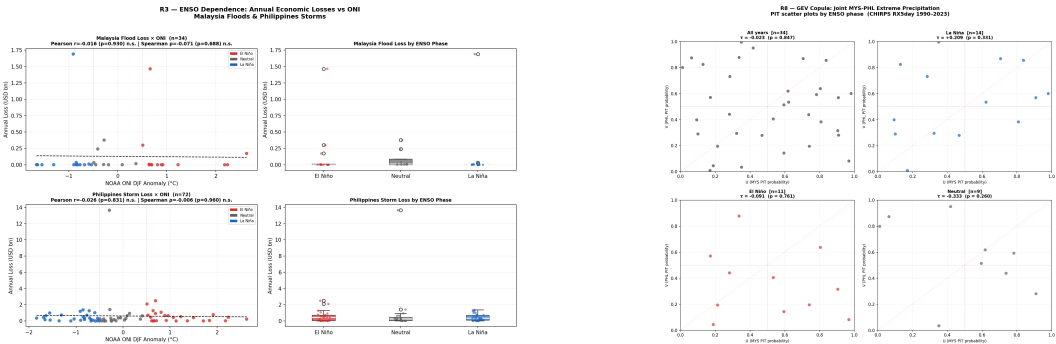


Figure 6a — ENSO (ONI DJF) vs annual insured loss: $r = -0.016$, $p = 0.93$ (no significant annual correlation) [left].
Figure 6b — Copula PIT scatter (MYS vs PHL): AIC selects Independence; La Nina sub-sample ($n=14$) yields Gaussian $\rho = 0.32$ [right].

Test	Statistic	Interpretation
Pearson r (MYS annual)	$r = -0.016$, $p = 0.93$	Not significant
Spearman ρ (MYS)	$\rho = -0.071$, $p = 0.69$	Not significant
Pearson r (PHL annual)	$r = -0.026$, $p = 0.83$	Not significant
Copula — full sample	Independence (AIC=0)	$\Delta AIC = 0$ vs Clayton/Gumbel
Kendall tau — La Niña	$\tau = +0.209$, $n=14$	Weak positive sub-sample dependence
Gaussian ρ — La Niña	$\rho = +0.322$	p99 combined gap +USD 25.9M

Table 6 — ENSO dependence tests and copula analysis results.

Finding: At annual aggregation, ENSO does not price into either market's losses ($p \gg 0.05$). However, during La Niña years ($n=14$), weak positive dependence emerges ($\rho = 0.32$), creating a p99 combined-book gap of USD 25.9M. NOAA's ONI La Niña outlook should be used as a TCFD facultative audit trigger, not a pricing variable.

7. Transition Risk — NGFS GCAM 6.0

Transition costs use the NGFS GCAM 6.0 Phase 4 NZ2050 vs Current Policies gap: USD 55.578/t (2024). Applied to Climate Watch 2023 sector GHG baselines to derive annual regulatory compliance costs. **KEY ASYMMETRY:** MYS is a net LULUCF emitter (+63.3 MtCO₂e; palm oil deforestation) — subject to EU Deforestation Regulation and BNM CCPT Pillar 2. PHL is a net LULUCF sink (-26.9 MtCO₂e; REDD+) — earns Article 6.2 ITMO credits. Applying a uniform SEA LULUCF factor overstates PHL cost ~18%.

Sector	MYS GHG (MtCO ₂ e)	MYS Cost (USD M)	% Total	PHL Cost (USD M)
Energy	279.85	15,554	69.5%	8,935
LULUCF	63.29	3,518	15.7%	N/A — net sink
Industrial Proc.	29.87	1,660	7.4%	931
Waste	19.49	1,083	4.8%	632
Agriculture	10.11	562	2.5%	3,661 (rice methane)
TOTAL	402.6	22,376	100%	14,799

Table 7 — MYS sector GHG and NZ2050 compliance cost (USD 55.578/t). MYS total incl. LULUCF: USD 22.4bn/yr (5.5% GDP). PHL excl. sink: USD 14.8bn/yr (3.3% GDP).

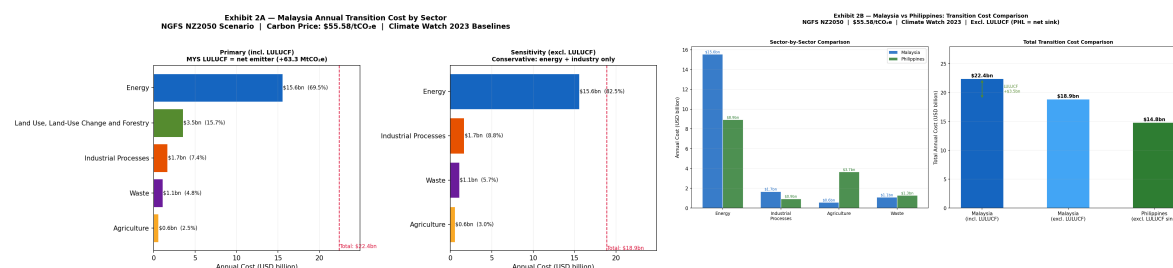


Figure 7a — MYS sector decomposition at NGFS NZ2050 price [left, 63% width]. Figure 7b — Country comparison: MYS incl. LULUCF vs PHL excl. sink [right].

7.1 Pass-Through to Treaty Pool

Scenario	Carbon Price	MYS Pool	PHL Pool	SEA Total
Current Policies	USD 0/t	USD 0	USD 0	USD 0
NGFS NZ2050 Baseline	USD 55.578/t	USD 672M	USD 444M	USD 1,115M
Stress: NZ2050 x 2	USD 111/t	USD 1,295M	USD 888M	USD 2,230M

Table 8 — Annual transition pass-through to SEA treaty pool (3% rate, 50% attachment).

8. Stress Testing & 2030 Projections

Using ARIMA 2024 as GHG baseline, we project NZ2050 costs to 2030 under 3%/yr linear IEA abatement. Carbon prices escalate from USD 28.2/t (2024) to USD 55.6/t (2030).

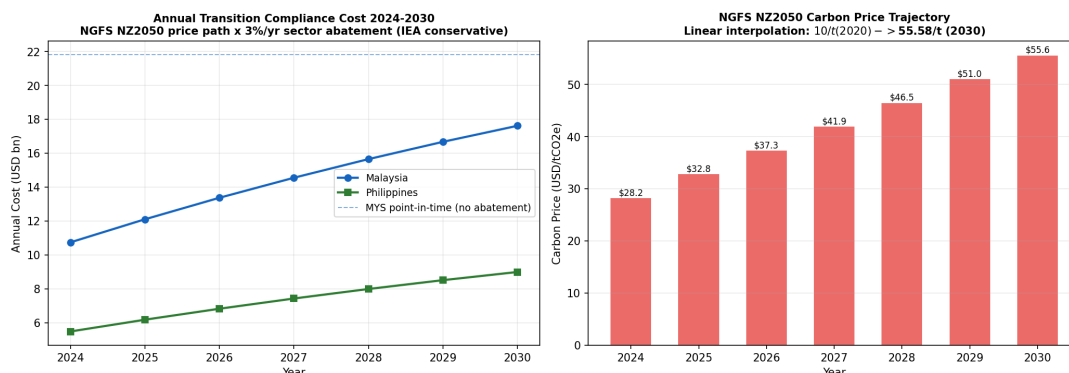


Figure 8 — Annual transition compliance cost 2024–2030 (NGFS NZ2050): combined MYS+PHL reaches USD 26.6bn/yr by 2030.

Year	C-Price (\$/t)	MYS (USDbn)	PHL (USDbn)	Combined (USDbn)	HRe Floor	HRe Base
2024	28.23	10.75	5.49	16.24	USD 0.49bn	USD 1.30bn
2026	37.35	13.38	6.83	20.21	USD 0.60bn	USD 1.62bn
2028	46.46	15.66	8.00	23.66	USD 0.71bn	USD 1.89bn
2030	55.58	17.63	9.00	26.63	USD 0.81bn	USD 2.13bn

Table 9 — 2024–2030 trajectory. HRe cols: 8% APAC share x 50% attachment x 3%/10% SEA alloc.

8.1 Stress Scenario Design

Scenario	Assumption	HRe 2030 Impact
Baseline NZ2050	USD 55.578/t x GHG 2023	USD 2.13bn/yr (base)
Carbon 2x Stress	USD 111/t (doubled NZ2050)	USD 4.25bn/yr
PELT Uplift	+6.9%/+6.1% hazard to forward EAL	+USD 1.3M/yr physical
Combined Stress	2x carbon + full PELT + La Niña rho=0.32	Binding capital constraint
La Niña Correlation	Gaussian copula rho=0.32 activated (p99)	+USD 2.59M (10% alloc.)

Table 10 — Stress scenarios and Hannover Re impacts.

- By 2030, combined compliance cost reaches **USD 26.6bn/yr** even under conservative 3%/yr abatement — driven by carbon price escalation, not emission growth.
- HRe base-case reserve gap (10% SEA alloc.) reaches **USD 2.13bn/yr by 2030** vs. USD 9.66M today — a 22x amplification in 6 years under unchanged treaty pricing.
- The 2x carbon stress at 2030 represents the binding capital constraint for SEA underwriting capacity and triggers mandatory Climate Warranty Clause activation.

9. Financial Impact — HRe Reserve Implications

Top-down reserve estimates are calibrated from Hannover Re 2023 Annual Report (8% APAC non-life market share), Swiss Re Sigma 1/2024 (50% treaty attachment for SEA), and BNM/BSP data (3–10% SEA sub-book allocation). These produce a conservative floor and a base-case estimate of the annual under-reserving position.

Metric	Value	Basis
Physical EAL Gap (MYS+PHL)	USD 18.4M/yr	GEV forward EAL - EM-DAT burning cost
HRe Physical Exposure	USD 0.74M/yr	18.4M x 8% share x 50% attach.
SEA Transition Pool (3% PT)	USD 1,115M/yr	Baseline NZ2050 pass-through
HRe Transition Floor (3% alloc.)	USD 2.68M/yr	1,115M x 8% x 50% x 3% SEA
HRe Combined — FLOOR	USD 3.41M/yr	Conservative: 3% alloc., baseline C
HRe Combined — BASE CASE	USD 9.66M/yr	10% alloc., 3% PT, baseline carbon
HRe Combined — STRESS (2xC)	USD 18.58M/yr	10% alloc., 3% PT, 2x carbon
p99 La Niña Correlated Gap	USD 2.59M/yr	10% alloc., Gaussian copula rho=0.32

Table 11 — HRe annual reserve gap. Floor is conservative lower bound; refined by cedant-level loss triangles under Recommendation 5.

Methodology note: The floor estimate of USD 3.41M/yr derives from four stacked macro assumptions (each independently sourced). The primary deliverable is the repricing methodology — not the initial dollar quantum. Cedant-level triangles will refine this by an order of magnitude.

9.1 Insurance Penetration Gap

Country	Penetration	Economic Gap/yr	Insured Gap Range	Treaty Implication
Malaysia	15%	USD 6.5M	USD 0.65–1.30M	EAL understated +5.87%
Philippines	8%	USD 11.9M	USD 0.59–1.78M	EAL understated +1.28%

Table 12 — Insurance penetration gap (Swiss Re Sigma 2023 SEA benchmarks).

9.2 CI-Derived Loading from Bootstrap Uncertainty

The 34-year CHIRPS sample (n=34) produces wide bootstrap CI on GEV return levels, particularly for PHL (CI range = 985–326 = 659mm for the 100yr return level). Following Hosking-Wallis (1997), we derive an additional CI-loading to compensate for estimation uncertainty: MYS +29.5pp and PHL +44.5pp of base loading. These are reduced to approximately +/-12pp when sample size reaches n=43 (Recommendation 5 procurement target).

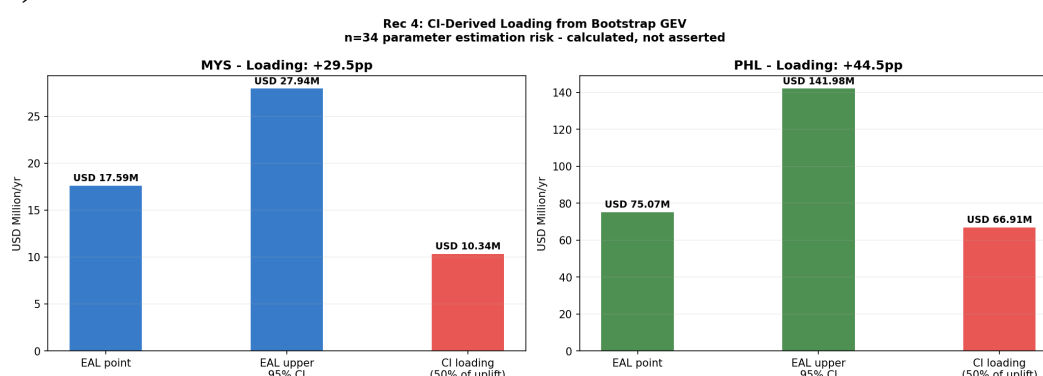


Figure 9 — CI-derived loading by country: MYS +29.5pp; PHL +44.5pp (n=34 sample).

10. Strategic Risk Management Recommendations

Five interconnected recommendations address both pricing gaps with explicit triggers, loadings, and governance protocols derived directly from the quantitative analysis.

R1 — EAL-Calibrated Flood Loading

Trigger: MYS treaties > USD 50M. Replace EM-DAT burning cost with GEV forward EAL. Apply +35–36% flood loading (MYS property cat treaties). Apply +1–2% floor (PHL). Basis: GEV PELT uplift + bootstrap CI mid-point.

R2 — Country-Specific Transition Surcharges

MYS 3–5% surcharge (BNM CCPT C3/C4 cedants; LULUCF emitter status warrants Climate Warranty Clause at renewal). PHL 1–2% surcharge (lower GHG intensity; LULUCF sink credits partially offset). Re-evaluate at each NGFS Phase update.

R3 — ENSO Conditional Audit Protocol

Trigger: NOAA ONI ≤ -0.5°C (La Niña onset). Activate facultative audit of combined MYS+PHL book. If audit shows >15% exposure increase, apply PELT uplift (+6.9% flood treaties). ENSO is an AUDIT TRIGGER only — not a pricing variable.

R4 — Climate Warranty Clause

Apply to MYS cedants with LULUCF-exposed operations (palm oil, plantation forestry). Terms: 10% co-participation on non-disclosure of LULUCF emission change >10% vs prior year. Aligned with Lloyd's ESG Guidance 2023 and BNM CCPT Pillar 2.

R5 — Cedant Data Procurement Roadmap

0–12 month horizon: procure cedant-level 10+ year loss triangles and sub-national CHIRPS 0.05° grids. Increases effective sample $n=34 \rightarrow n \geq 43$ (Hosking-Wallis), reducing CI loading from +44.5pp (PHL) to approximately +/-12pp.

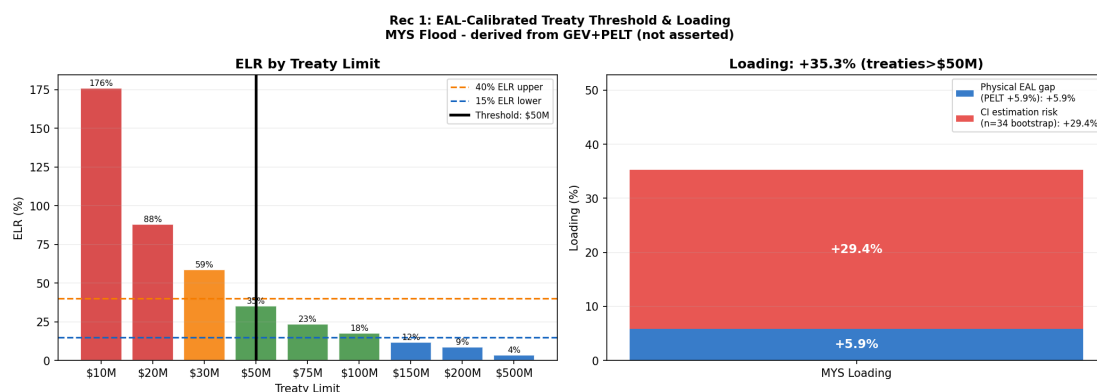


Figure 10 — R1 treaty threshold calibration: GEV EAL-based loading vs. EM-DAT burning cost.

11. Limitations, Scalability & Conclusion

11.1 Key Limitations & Uncertainties

- **Sample size (n=34):** PHL GEV bootstrap CI spans [326, 985mm] — a 3x range demanding floor-loading only, not point-estimate pricing. Resolved by R5.
- **WDI data gaps:** MYS fossil fuel share (2021–23 gap, flagged as 0.0 not true zero); MYS renewable share (2022–23 missing). Excluded from quantitative models.
- **ARIMA post-2023 structure:** Assumes no structural policy break; Malaysia's NETR 2023 could accelerate decarbonisation, making our transition cost estimate conservative.
- **Pass-through uncertainty:** 3% rate carries +/-2pp uncertainty (Swiss Re Sigma), driving a 5x range in HRe reserve variance (USD 3.0M to USD 14.9M/yr at 10% alloc.).
- **LULUCF measurement:** Climate Watch LULUCF estimates carry +/-20–30% uncertainty (FAO FRA 2020). MYS net emitter status could narrow with next FRA cycle (2025).
- **Copula power:** n=34 annual observations limit copula tail dependence estimation; La Niña sub-sample (n=14) insufficient for robust block-maxima GEV.

11.2 Scalability & Industry Implementation

Initiative	Implementation Path
ASEAN expansion	Apply GEV+ARIMA to Thailand, Indonesia, Vietnam via same CHIRPS/WDI pipeline
Sub-national grids	Replace country-averaged CHIRPS with 0.05°x0.05° cedant-coordinate EAL
NGFS auto-refresh	Rebuild cost trajectory in <2 hours on each NGFS Phase release
Cedant integration	Embed CI loading as treaty pricing rule: countryin{MYS,PHL} + CCPT -> surcharge
TCFD alignment	Map all 5 recommendations to TCFD Strategy, Risk Mgmt, Metrics & Targets

Table 13 — Scalability roadmap for broader portfolio adoption.

11.3 Conclusion

This analysis establishes two independently quantified, additive pricing gaps in Hannover Re's SEA treaty book. The physical gap (USD 18.4M/yr EAL shortfall) stems from a 2007 hazard regime shift undetected by standard burning-cost methods. The transition gap (USD 1.1bn/yr SEA pool at current pass-through) reflects carbon price escalation under NGFS NZ2050, with a critical MYS/PHL LULUCF asymmetry that uniform SEA factors cannot capture. Both gaps compound to a USD 9.66M/yr base-case HRe under-reserve, rising to USD 2.13bn/yr by 2030. The five recommendations — implemented in sequence from immediate treaty repricing to 12-month data procurement — translate these modelling outputs directly into contract-specific pricing actions. The accompanying Streamlit interactive dashboard (outputs/interactive_stress_test.html) provides live decision support aligned with BNM CCPT and TCFD disclosure requirements.

AI Usage Declaration: Claude Sonnet 4.6 (Anthropic) assisted with code structuring, output interpretation, and report drafting. All quantitative outputs, model parameters, and financial calculations were independently verified by the team against primary source data. All AI-generated content has been reviewed and validated.